MINHAZUL ISLAM

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School of Software Technology, Zhejiang University, Ningbo, Zhejiang, China

Research Interests

I study humanAI collaboration in high-stakes domains, including mental-health peer support, health communication, and expert decision-making. My research focuses on trustworthy language-model systemsgrounding, hallucination detection, evaluation, and human oversightand on AI-mediated communication that supports human skills while preserving agency, judgment, and voice.

Education

- **Zhejiang University** — *Sept. 2023 – Dec. 2026 (Expected)*
  M.Eng. in Industrial Design Engineering — Ningbo, China
  - **Thesis:** *BEPSBot: A Dual-Mode Writing Assistant for Online Bipolar Disorder Peer Support Communities.* Advisor: Prof. Mengru Xue.
- **Yunnan University** — *Sept. 2019 – June 2023*
  B.Eng. in Computer Science and Technology — Kunming, China
  - **GPA:** 3.6/4.0
  - **Thesis:** *Product Recommendation System Based on a Collaborative Filtering Algorithm.* Evaluated matrix-factorization and similarity-based approaches on goodbooks-10k.

Publications & Presentations

Peer-Reviewed Conference Papers

[C.1] **Minhazul Islam**, Mengru Xue, and Tasnim Afra (2025). *Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 61–75. doi:10.1007/978-3-032-02534-0_8

[C.2] Tasnim Afra, Mengru Xue, and **Minhazul Islam** (2025). *Enhancing Biofeedback Interventions for Depression and Anxiety Through Entertainment Computing: A Systematic Review*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 37–51. doi:10.1007/978-3-032-02534-0_6

Conference Presentations & Non-Archival Papers

[P.1] **Minhazul Islam**, Mengru Xue, and Tasnim Afra (2026). *BEPSBot: Draft-Grounded AI Writing Assistance for Bipolar-Related Online Peer Support*. Oral presentation, *HHME 2026 PCC*, Zhejiang University, China. [conference page]

[P.2] Tasnim Afra, Mengru Xue, and **Minhazul Islam** (2026). *How Should Voice Agents Respond to Anger and Sadness? A Comparison of Empathic Response, Affect-Neutral Acknowledgment, and Cognitive Reappraisal*. Oral presentation, *HHME 2026 PCC*, Zhejiang University, China. [conference page]

Research Experience

- **Zhejiang University — Graduate Research Assistant** — *Sept. 2023 – 2026*
  Advisor: Prof. Mengru Xue — Ningbo, China
  - **Human–AI collaboration for mental-health peer support (BEPSBot).** Designed a draft-grounded generative writing assistant that preserves a peer-support provider's semantic content while using AI to improve expression, studying how generative assistance can increase utility without displacing human authorship.
  - Conducted a controlled study with 24 peer-support providers; draft-grounded generation increased recommendation adoption from 33.3% to 75.6%, achieved a mean SUS score of 86.98, and preserved improvements in user confidence and satisfaction.
  - Built a dual-mode pipeline combining contextual RoBERTa assessment, sentence-embedding retrieval with ChromaDB, retrieval-grounded LLM generation, strategy-constrained rewriting, and pre/post-generation safety screening.
  - **Clinician-centered AI for bipolar-disorder monitoring.** Conducted semi-structured interviews with seven mental-health professionals and thematic analysis to identify physiological, behavioral, and psychosocial signals relevant to longitudinal bipolar-disorder monitoring; translated findings into design requirements for clinician-centered multimodal AI decision support [C.1].
  - **Research through design.** Designed and prototyped SPROUT, a shape-changing meditative artifact targeting fear of failure in students, combining embedded sensing and actuation with a growth metaphor to explore how tangible interaction can support reflective practice.

Applied Research & Engineering Experience

- **Synexia AI** *2025 – Present*
  AI Infrastructure & Systems Engineer Hangzhou, China
  - Architected two production agentic-AI systems — *Zhanlu*, a governed multi-agent runtime, and *EDIA*, a 17-node LangGraph decision-intelligence system — supporting complex human decision workflows through structured planning, tool use, verification, and evidence-grounded generation.
  - Built agent-runtime mechanisms including plan-first execution, sub-agent workflows, goal contracts and bounded self-correction, LLM-as-a-judge evaluation, tool/skill/MCP gateways, and context-isolation controls for reliable multi-agent behavior.
  - Reduced EDIA SQL round trips from 77 to 2–5 and retrieved rows from roughly 180,000 to 50–200 through intent-driven query planning and deterministic metric pre-computation, reducing end-to-end latency from 90–478 s to 30–45 s while removing a class of numerical hallucinations.
  - Built verifiable-output infrastructure with golden-eval pipelines, citation-grounding checks, hybrid SQL/vector retrieval (ChromaDB, BAAI/bge-m3), and deterministic validation gates that block unsupported claims from reaching users.
  - Designed a privacy-preserving dual-LLM architecture in which an on-premise Qwen model reads and sanitizes sensitive enterprise records before external reasoning is allowed, with external routing blocked unless privacy conditions are satisfied.
- **Piliving** *June 2024 – Sept. 2024*
  Technical Product Manager (concurrent, part-time) Ningbo, China
  - Analyzed 100,000+ user forum posts and comments using text mining and qualitative categorization to identify recurring product failure modes and translate user evidence into design requirements and prototype revisions.
- **ProFabx** *June 2024 – Sept. 2024*
  Software Engineer (concurrent, part-time) Ningbo, China
  - Automated parametric CAD workflows through the Autodesk Inventor API in Python, reducing manual modeling time by roughly 40% and supporting rapid design-space exploration through a browser-based parameter interface.

Selected Technical Projects

- **InclusiveVision — Assistive Smart Glasses for Visually Impaired Users** *Dec. 2023*
  Zhejiang University []
  - Designed and iteratively prototyped an assistive wearable providing real-time auditory obstacle feedback, integrating ultrasonic sensing, embedded interaction, and 3D-printed form-factor iterations for wearability.

Honors & Awards

- **United Nations Technology Bank —** Global Youth Talent Program, selected internationally among young designers *2023*

Technical Skills

- **Research Methods:** Controlled user studies, semi-structured interviews, thematic and qualitative coding, research through design, text mining of large social corpora, benchmarking and ablation, usability and user-facing evaluation
- **Human–AI & Interactive Systems:** Human–AI collaboration, agentic decision support, interactive dashboard design, human-centered prototyping, React.js, D3.js, Recharts
- **Multi-Agent & Agentic AI:** LangGraph (17-node FSM), agent harness/runtime design, plan-first execution, sub-agent workflows, goal contracts, bounded self-correction, tracing and observability layers, MCP, tool/skill gateways, LLM-as-a-judge evaluation
- **Context Engineering & Retrieval:** Context isolation and spill management, hybrid SQL/vector RAG, ChromaDB, BAAI/bge-m3, context trimming, deterministic guardrails, provenance and output validation
- **AI Infrastructure & LLMOps:** FastAPI, SSE streaming, Docker (14-service stack), vLLM, Qwen3-27B, Ollama, PostgreSQL, Redis, MinIO, multi-provider routing, dual-LLM security gates, Prometheus/Grafana, systemd, Nginx
- **Programming & ML:** Python (asyncio), C/C++, SQL, JavaScript, PyTorch, Transformers, scikit-learn, XGBoost, LoRA/QLoRA/PEFT, Git, GitHub Actions CI/CD
- **Languages:** Bengali (native), English (fluent), Mandarin Chinese (conversational)